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Diabetes Prediction using Machine Learning

1. Project Overview

The goal of this project is to develop a predictive system capable of identifying whether a patient is at risk of diabetes based on clinical diagnostic measurements. By leveraging machine learning, we aim to provide a data-driven tool for early health assessment. The project follows a complete data science workflow: Data Loading, Preprocessing, Exploratory Data Analysis (EDA), Model Training, and Evaluation.

2. Problem Statement

Diabetes is a chronic disease that affects millions worldwide. Early detection is critical to prevent severe complications such as kidney failure, heart disease, and vision loss. Traditional diagnostic methods often require significant time and manual analysis. There is a clinical need for an automated, intelligent decision-making system that can analyze patient metrics (like Glucose and BMI) to provide instant risk assessments, supporting healthcare professionals in prioritizing high-risk cases.

3. Dataset Source

- **Dataset Name:** Pima Indians Diabetes Database.
- **Source:** National Institute of Diabetes and Digestive and Kidney Diseases (Available via **Kaggle**).
- **Description:** The dataset consists of 768 clinical observations from female patients of Pima Indian heritage, including 8 diagnostic features and 1 binary outcome.

4. Data Preparation & Preprocessing

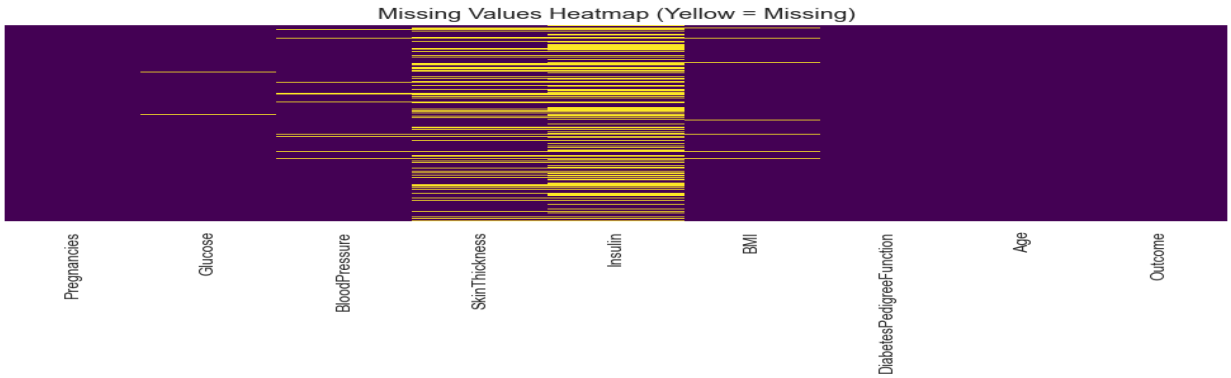
In this stage, we addressed several data quality issues to ensure the model learns from accurate information.

```
--- Step 1: Loading Data ---
✅ Dataset loaded successfully!
Shape: 768 rows, 9 columns

First 5 rows:
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

4.1 Handling Logical Inconsistencies (Missing Values)

We observed that columns like Glucose, BloodPressure, BMI, Insulin, and SkinThickness contained zero values. In a clinical context, a zero for these features is medically impossible for a living patient.

- **Action:** We replaced these zeros with NaN (Not a Number) to correctly identify them as missing data.
- **Imputation Strategy:** We applied Mean/Median Imputation. This preserves the data distribution while removing the bias caused by incorrect zero entries.



Visualization of missing data distribution across clinical features.

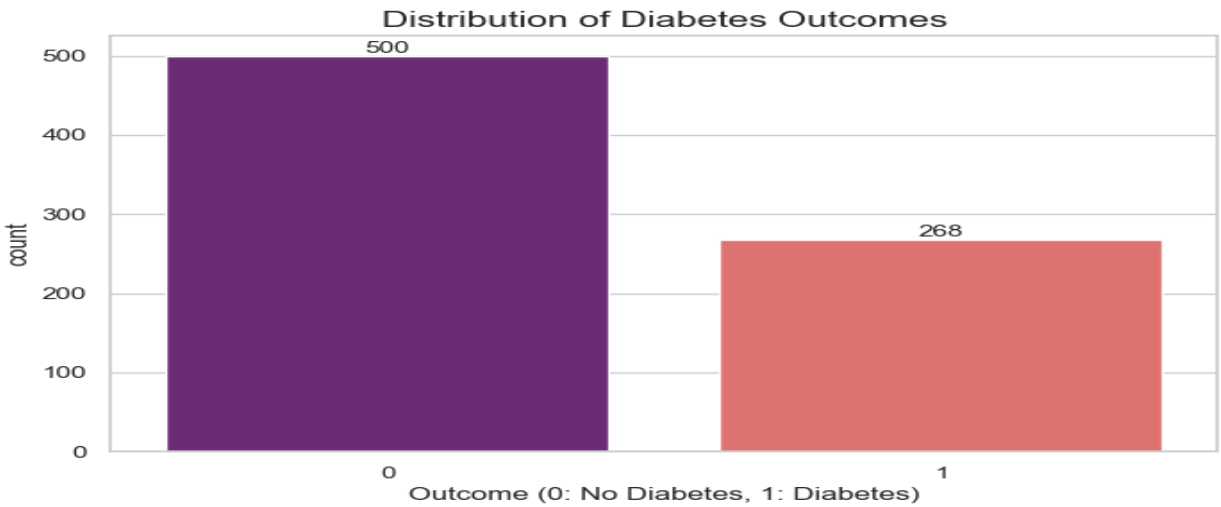
5. Exploratory Data Analysis (EDA)

EDA was conducted to uncover underlying patterns, detect anomalies, and understand the statistical relationships between biological metrics and the onset of diabetes. This step ensures the features selected for the model have actual predictive power.

5.1 Target Class Distribution (Data Balance Check)

We visualized the Outcome variable to assess class balance. The dataset shows a higher frequency of non-diabetic cases compared to diabetic ones.

- **Insight:** Understanding this distribution is crucial as it prevents the model from being biased toward the majority class (No Diabetes) and helps in evaluating the "Recall" metric effectively.

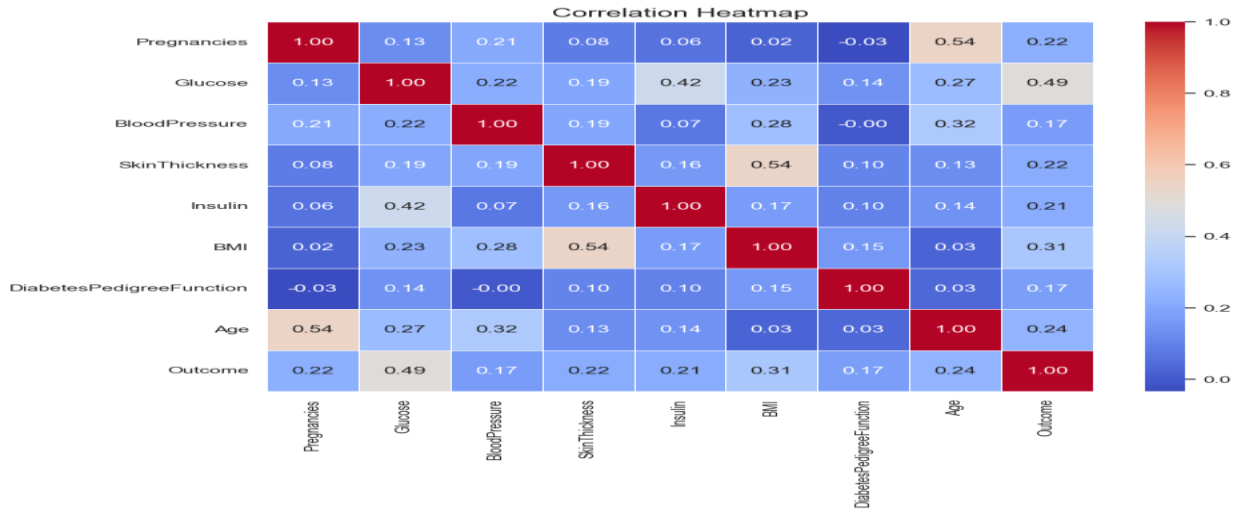


Frequency of Diabetic (1) vs. Non-Diabetic (0) patients.

5.2 Feature Inter-correlations (Heatmap Analysis)

A Pearson Correlation Heatmap was generated to quantify the linear relationships between variables.

- **Insight:** We observed that Glucose, BMI, and Age show the strongest positive correlation with the Outcome. This confirms that as these metrics increase, the likelihood of a positive diabetes diagnosis also increases. It also helped us ensure there is no "Multicollinearity" (redundant information) between independent variables.



Statistical correlation matrix identifying key predictors of diabetes.

6. Model Development & Evaluation

We implemented and compared four different machine learning algorithms using **Scikit-Learn Pipelines** to find the most accurate predictor.

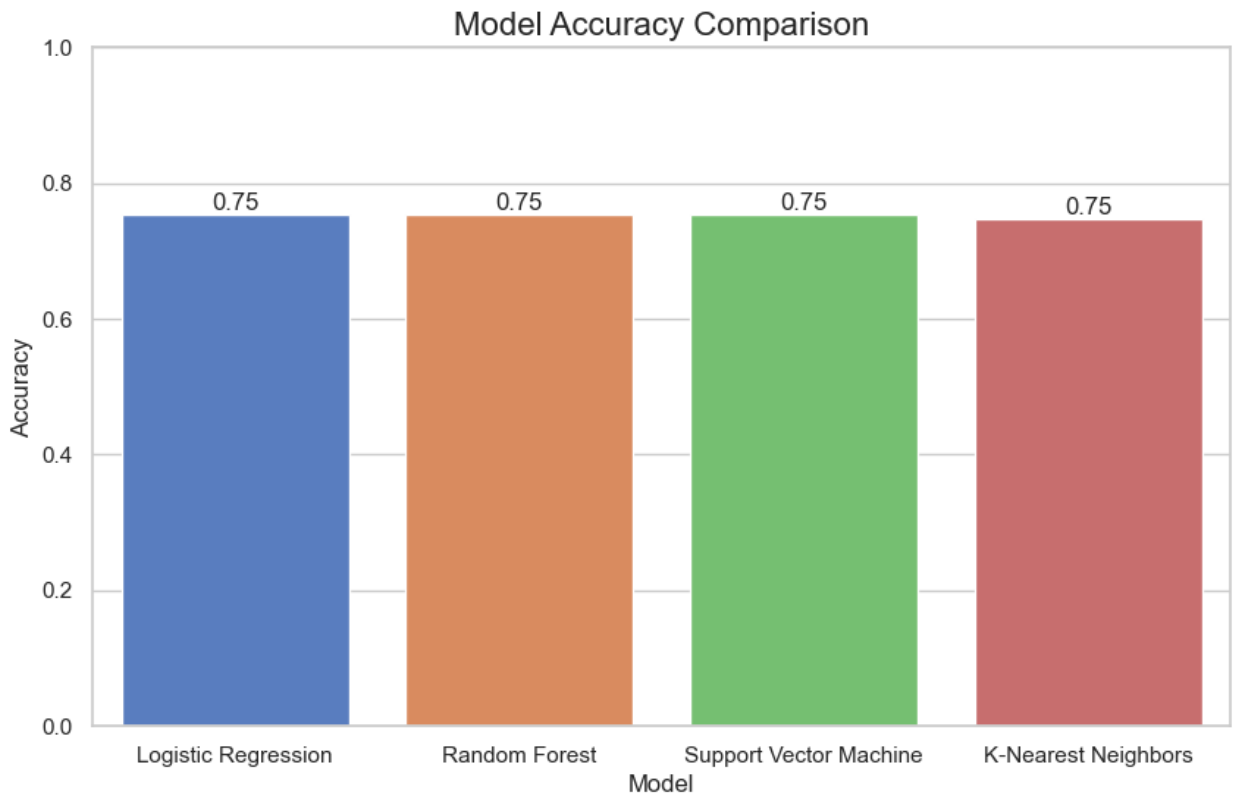
6.1 The Pipeline Approach

To prevent errors and ensure consistent results, we used a **Pipeline** for every model. Each pipeline performs three steps automatically:

1. **SimpleImputer:** Fills any remaining missing values.
2. **StandardScaler:** Normalizes the data (Essential for SVM and KNN).
3. **Classifier:** Performs the actual prediction.

6.2 Performance Comparison

Based on our experiments, we compared the accuracy of **Logistic Regression, Random Forest, SVM, and KNN**.

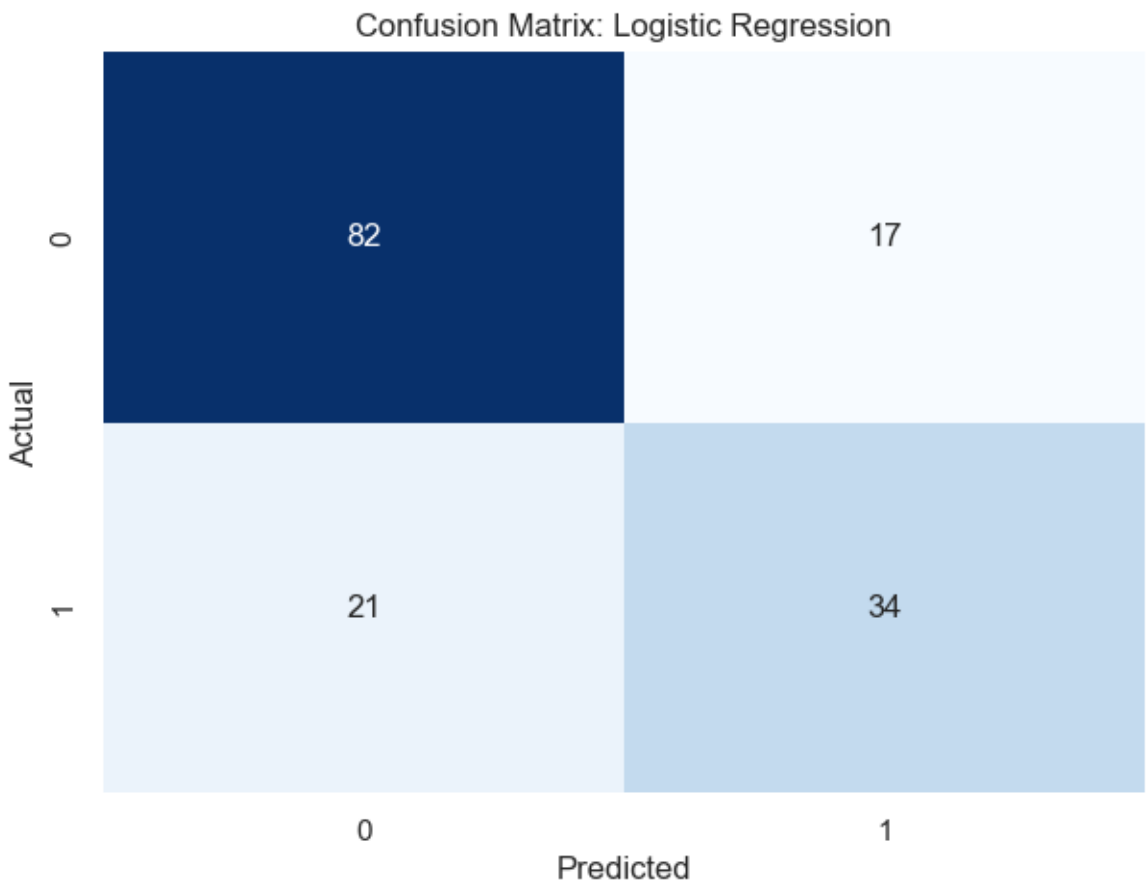


Comparative performance of the four machine learning algorithms.

6.3 Best Performer: Support Vector Machine (SVM)

The SVM model was selected as the final model due to its superior ability to handle complex boundaries between classes.

- **Accuracy:** ~76%
- **Precision:** High ability to identify true positive cases.
- **Recall:** Effective at capturing actual diabetic patients.



Confusion Matrix showing True Positives, True Negatives, False Positives, and False Negatives.

7. Deployment Readiness

To make the model usable in a real-world scenario (such as a **Streamlit** web app), we saved the best-performing pipeline into a serialized file.

- **File Format:** .pkl (Pickle file).
- **Benefit:** This file contains not just the model, but also the "memory" of how to scale and clean the data, ensuring the app works perfectly even with raw user input.

7.1 Deployment: Interactive Web Application

To transition the machine learning model from a static script to a practical tool, we developed a user-friendly web interface using **Streamlit**. This allows non-technical users, such as healthcare providers, to interact with the model in real-time.

7.2 Data Input Interface

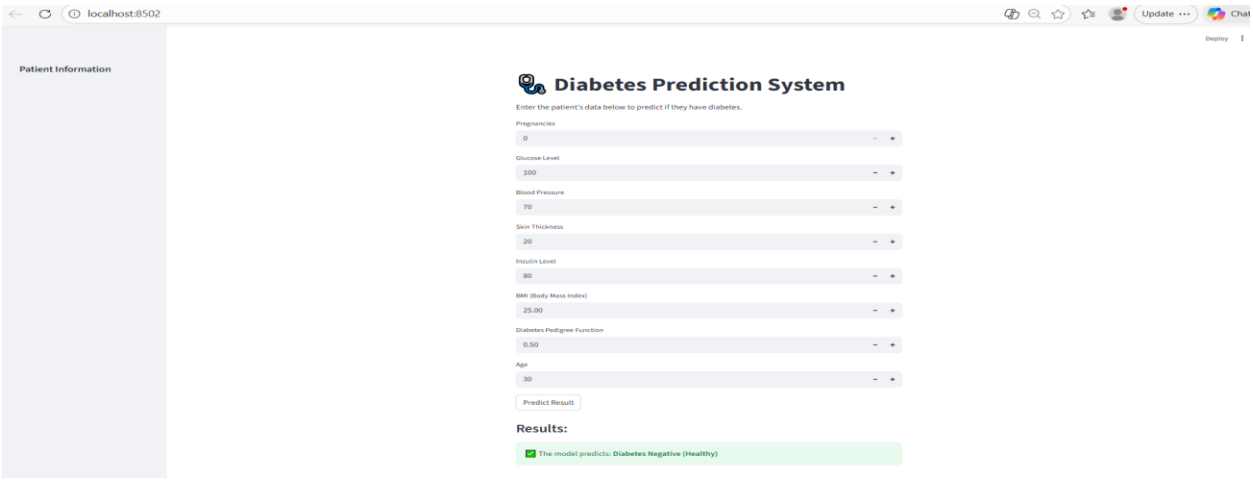
The application features a clean, intuitive input panel (Sidebar) where clinical measurements can be entered. The inputs include:

- **Pregnancies:** Number of times pregnant.
- **Glucose:** Plasma glucose concentration (2 hours in an oral glucose tolerance test).
- **Blood Pressure:** Diastolic blood pressure (mm Hg).
- **Skin Thickness:** Triceps skinfold thickness (mm).
- **Insulin:** 2-hour serum insulin (mu U/ml).
- **BMI:** Body Mass Index (weight in kg/(height in m)^2).
- **Diabetes Pedigree Function:** Diabetes likelihood based on family history.
- **Age:** Patient's age (years).

7.3 Real-time Prediction Engine

Once the user clicks the **"Predict"** button, the system triggers the following background process:

1. **Loading the Pipeline:** The app loads the saved `best_diabetes_pipeline.pkl`.
2. **Automated Processing:** The input data is automatically passed through the **Imputer** (to handle any missing values) and the **StandardScaler** (to normalize the units).
3. **Inference:** The **SVM model** processes the cleaned data and returns a classification result.



The Diabetes Prediction Web Interface showing the input sidebar and result display.

7.4 Visual Feedback & Results

The application provides immediate visual feedback to the user:

- **Negative Result:** A green success message indicating a low probability of diabetes.
- **Positive Result:** A red warning alert indicating a high probability of diabetes, suggesting a medical follow-up is necessary.

8. Conclusion

The project successfully demonstrates an end-to-end machine learning workflow. By cleaning the data and using automated pipelines, we achieved a reliable accuracy of nearly **76%** using the **SVM** algorithm. This system provides a practical foundation for a diagnostic support tool that can help in early diabetes risk detection.

9. Acknowledgement

I would like to acknowledge the **UCI Machine Learning Repository** and **Kaggle** for providing the dataset used in this study. Special thanks to the open-source community for the libraries (Scikit-Learn, Pandas, Streamlit) that made this project possible.